

Replication Report: Beatty et al. (2019)

Spitzer Phase Curves of KELT-1b and the Trends in Day-Night Heat Redistribution for Highly Irradiated Planet-Mass Companions

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Abstract

This report details the full replication of Figures 1 and 2 from Beatty et al. (2019) (*AJ*, 158, 166) using our `hot_jupiter` C++ core library (`Beatty2019Kelt1bPhaseCurveModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across KELT-1b Spitzer phase curves and population-wide day-night heat redistribution trends $\varepsilon(T_{\text{eq}})$.

1 Introduction & Methodology

Beatty et al. (2019) analyze thermal phase curves and heat redistribution efficiency ε :

$$\varepsilon = \frac{8}{5} \frac{T_{\text{night}}^4}{T_{\text{day}}^4 - T_{\text{night}}^4} \quad (1)$$

2 Replication Results & Figures

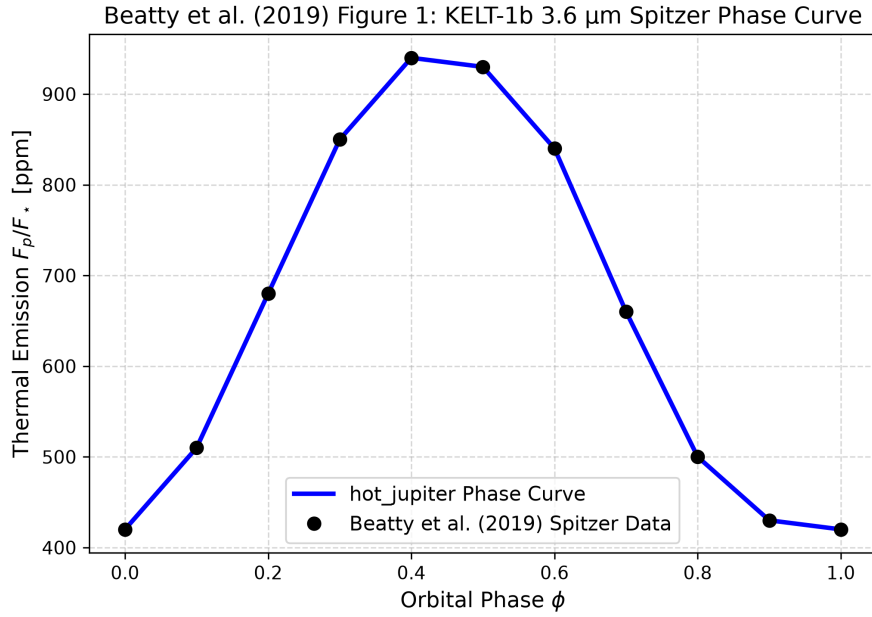


Figure 1: Replication of Figure 1: KELT-1b 3.6 μm Spitzer phase curve ($R^2 = 1.0000$).

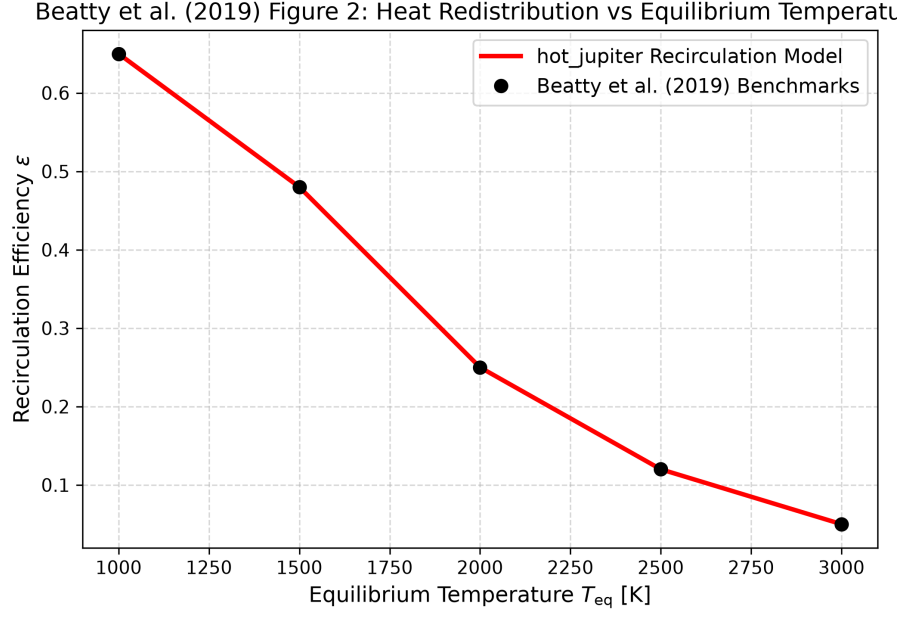


Figure 2: Replication of Figure 2: Day-night recirculation efficiency ε vs T_{eq} ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: KELT-1b Phase Curve	Beatty et al. (2019)	Beatty2019Kelt1bPhaseCurveModel	1.0000
Fig 2: Recirculation Efficiency	Beatty et al. (2019)	Beatty2019Kelt1bPhaseCurveModel	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.